LOG-CONCAVITY OF INVERSE Z-POLYNOMIALS OF BOSE–BURTON MATROIDS

KNOWLEDGE SENSE

Abstract. The inverse Z-polynomial $Y_M(t)$ of a matroid M, introduced by Ferroni, Matherne, Stevens and Vecchi and studied by Gao, Ruan and Xie, has nonnegative, symmetric and unimodal coefficients, and its coefficients are conjectured to be log-concave. We prove this conjecture, in strict form, for every Bose–Burton matroid $B_{r,s}(q)$, that is, for the restriction of the projective geometry $\mathrm{PG}(r-1,q)$ to the complement of a projective subspace of rank s, for every prime power q, every $r \geq 2$ and every $1 \leq s \leq r-1$. The family contains the affine geometries $\mathrm{AG}(r-1,q)$. We deduce this from a stronger statement: after division by the coefficients of the inverse Z-polynomial of $\mathrm{PG}(r-1,q)$, the coefficient sequence of $Y_{B_{r,s}(q)}$ is positive and concave. The proof rests on an explicit expansion of the normalized coefficients as a positive combination of products of q-Pochhammer type; concavity is proved by comparing the weighted second differences of adjacent terms of this expansion at the boundaries of their supports.

1. Introduction

Let M be a loopless matroid on a ground set E, with lattice of flats $L(M)$. The Kazhdan–Lusztig polynomial $P_M(t)$ of Elias, Proudfoot and Wakefield [6] and the Z-polynomial $Z_M(t)$ of Proudfoot, Xu and Young [12] have inverse counterparts in the incidence algebra of $L(M)$: the inverse Kazhdan–Lusztig polynomial $Q_M(t)$ of Gao and Xie [11], and the inverse Z-polynomial $Y_M(t)$. The latter first appeared, with a different sign convention, in the work of Ferroni, Matherne, Stevens and Vecchi [7]; its systematic study was begun by Gao, Ruan and Xie [10], and a deletion formula for it was given by Braden, Ferroni, Matherne and Nepal [3]. In the normalization of [10, (2)–(3)] and [3, Theorem 2.2],

$$Y_M(t) = \sum_{F \in L(M)} (-1)^{\operatorname{rk} M - \operatorname{rk} F}\, \mu(M/F)\, t^{\operatorname{rk} M - \operatorname{rk} F}\, Q_{M|F}(t), \tag{1.1}$$

where $\mu(M/F)$ is the Möbius invariant of the contraction M/F.

Gao, Ruan and Xie proved that $Y_M(t)$ has nonnegative coefficients [10, Proposition 2.1], using the nonnegativity of inverse Kazhdan–Lusztig polynomials established by Braden, Huh, Matherne, Proudfoot and Wang [4], and that $Y_M(t)$ is palindromic of degree $\operatorname{rk} M$ [10, Lemma 2.2]. They conjectured that its coefficients are unimodal [10, Conjecture 1.4] and, more strongly, that they form a log-concave sequence with no internal zeros [10, Conjecture 1.5], and they proved both statements for sparse paving matroids [10, Theorem 1.6]. Unimodality has

Date: 12 October 2026.

2020 *Mathematics Subject Classification.* Primary 05B35; Secondary 05A20, 05A30, 51E20.

Key words and phrases. Inverse Z-polynomial, Bose–Burton geometry, projective geometry, affine geometry, log-concavity, Gaussian binomial coefficients.

since been proved for all loopless matroids; see the manuscript [13]. For q-niform matroids, Gao, Li and Xie proved strong log-concavity of the inverse Z-polynomial [9, Corollary 3.9], and explicit formulas are known for further families, for instance fan matroids [8, Theorem 1.3]. Log-concavity in general remains open.

In this paper we prove log-concavity for the Bose–Burton matroids, a family of representable matroids that are not paving as soon as the rank is at least 5 (Remark 1.4). Let q be a prime power, let $r \geq 2$, let V be an r-dimensional vector space over $\mathbb{F}_q$, and let $W \subseteq V$ be a subspace of dimension s with $1 \leq s \leq r-1$. We write $\mathbb{P}(U)$ for the set of one-dimensional subspaces of a subspace $U \subseteq V$, and $B_{r,s}(q)$ for the simple matroid on $\mathbb{P}(V)\setminus\mathbb{P}(W)$ in which a set of points is independent if representative vectors are linearly independent. Thus $B_{r,s}(q)$ is the restriction of the projective geometry $\mathrm{PG}(r-1,q)$ to the complement of $\mathbb{P}(W)$; up to isomorphism it depends only on (r,s,q). These point sets are the extremal configurations in a theorem of Bose and Burton [2], and Cheng and Liu used them as the starting point of their construction of matroids with non-unimodal Kazhdan–Lusztig polynomials; they also computed P and Z for every $B_{r,s}(q)$ [5, Remark 2.7]. The family contains the affine geometries $\mathrm{AG}(r-1,q) = B_{r,r-1}(q)$, the complements $B_{r,1}(q)$ of a single point, and, for $r = 2$, the uniform matroids $U_{2,q}$.

Theorem 1.1. *Let q be a prime power and let r and s be integers with $r \geq 2$ and $1 \leq s \leq r-1$. Write $Y_{B_{r,s}(q)}(t) = \sum_{k=0}^{r} y_k t^k$. Then $y_k > 0$ and $y_k = y_{r-k}$ for $0 \leq k \leq r$, and*

$$y_k^2 > y_{k-1}\, y_{k+1} \qquad (1 \leq k \leq r-1).$$

In particular $Y_{B_{r,s}(q)}(t)$ has degree r, and its coefficients form a strictly log-concave sequence with no internal zeros.

Nonnegativity and symmetry hold for all matroids by [10, Proposition 2.1 and Lemma 2.2]; we include positivity and symmetry in the statement because the proof gives them directly. The new assertion is the strict log-concavity. It follows from a stronger statement about normalized coefficients. By a result of Gao, Li and Xie [9, Proposition 3.8] (see Lemma 2.2 below), the coefficient of t^k in the inverse Z-polynomial of $\mathrm{PG}(r-1,q)$ is

$$\pi_k = q^{\binom{k}{2}+\binom{r-k}{2}} \begin{bmatrix} r \\ k \end{bmatrix}_q \qquad (0 \leq k \leq r), \tag{1.2}$$

and this sequence is strictly log-concave. Put $z = q^{-1}$, $c = r - s$, and $(z;z)_m = \prod_{i=1}^{m}(1-z^i)$, with $(z;z)_0 = 1$. For integers $j \geq 0$ and $1 \leq k \leq r-1$ put

$$R_j(k) = \prod_{i=1}^{j} \bigl(1 - z^{k-i}\bigr)\bigl(1 - z^{r-k-i}\bigr) \tag{1.3}$$

and, for $0 \leq j \leq s$,

$$w_j = (1-z^c)\, z^{j(c+j+1)} \begin{bmatrix} s \\ j \end{bmatrix}_z \frac{(z;z)_{r-j-1}}{(z;z)_r\,(z;z)_j}. \tag{1.4}$$

Some factors of $R_j(k)$ have negative exponents when $j > \min(k, r-k)$; in that case the product also contains the factor $1 - z^0 = 0$, so $R_j(k) = 0$.

Theorem 1.2. *Let q, r, s and y_k be as in Theorem 1.1, and put $F_k = y_k/\pi_k$ for $0 \leq k \leq r$. Then the following hold.*

(i) $F_0 = F_r = \prod_{i=c}^{r-1}(1-z^i)$.

(ii) *For* $1 \le k \le r-1$,

$$F_k = \sum_{j=0}^{s} w_j \, R_j(k),$$

where $w_j > 0$ *for* $0 \le j \le s$, *and* $R_j(k) > 0$ *if* $j < \min(k, r-k)$ *while* $R_j(k) = 0$ *if* $j \ge \min(k, r-k)$.

(iii) *The sequence* $F_0, F_1, \dots, F_r$ *is positive, symmetric and concave:*

$$2F_k \ge F_{k-1} + F_{k+1} \qquad (1 \le k \le r-1),$$

with strict inequality for $k = 1$ *and* $k = r-1$.

For the full projective geometry, which would correspond to $s = 0$, the normalized sequence is identically one. Theorem 1.2 thus says that removing the points of a subspace multiplies the inverse Z-coefficients of $\mathrm{PG}(r-1, q)$ by a concave profile. Concavity of F is strictly stronger than log-concavity of F, and since π is strictly log-concave, Theorem 1.2 implies Theorem 1.1 (Section 6).

The expansion in Theorem 1.2(ii) has positive coefficients, but this alone does not give concavity: each product R_j is concave on the range where it is positive, which for $j \ge 1$ is $\{j+1, \dots, r-j-1\}$. If $j \ge 2$ and $r \ge 2j+2$, then at the index $k = j$ just to the left of this support we have $R_j(j-1) = R_j(j) = 0$, so the second difference $R_j(j+1) + R_j(j-1) - 2R_j(j) = R_j(j+1)$ is positive, and by symmetry the same happens at the index $k = r-j$ just to the right of the support. Positive combinations of log-concave or piecewise concave sequences need not be log-concave, so these positive second differences must be compensated. The proof in Section 5 compares the weighted second differences of adjacent products at the support boundary: at the index $k = j$, the positive second difference of $w_j R_j$ is outweighed by the negative second difference of $w_{j-1} R_{j-1}$, and the right boundary is handled by symmetry. This comparison uses $q \ge 2$, through the inequality $z \le 1/2$.

Corollary 1.3. *Conjecture 1.5 of* [10] *holds for every Bose–Burton matroid* $B_{r,s}(q)$ *with* $r \ge 2$ *and* $1 \le s \le r-1$.

Remark 1.4. Some small members of the family are sparse paving and are therefore covered by [10, Theorem 1.6]: for example $B_{2,1}(q) = U_{2,q}$ and $B_{3,2}(2) = \mathrm{AG}(2,2) = U_{3,4}$. For $r \ge 5$, however, $B_{r,s}(q)$ is not paving. Indeed, a 3-dimensional subspace $U \not\subseteq W$ meets W in dimension at most 2, so the set $\mathbb{P}(U) \setminus \mathbb{P}(W)$ has at least $(q^2+q+1) - (q+1) = q^2 \ge 4$ elements and rank 3. Any four of these points are dependent, so $B_{r,s}(q)$ has a circuit with at most $4 < r$ elements. Theorem 1.1 does not follow from real-rootedness either: by Example 3.4, $Y_{B_{2,1}(3)}(t) = 2 + 3t + 2t^2$, which has no real roots.

Inputs and organization. We use the following results from the literature: the characterization of Y_M by inversion in the incidence algebra [3, Section 2.2]; the formulas for P and Z of Bose–Burton matroids [5, Remark 2.7]; and the formula for the inverse Z-polynomial of a projective geometry [9, Proposition 3.8]. Section 2 recalls these results together with the description of the flats of $B_{r,s}(q)$. In Section 3 we combine them into an explicit signed formula for $Y_{B_{r,s}(q)}$ (Proposition 3.2). Section 4 converts this formula into the positive expansion of Theorem 1.2(i)–(ii), and Section 5 proves the concavity in Theorem 1.2(iii). Section 6 completes the proof of Theorem 1.1 and gives examples.

2. Preliminaries

2.1. **Gaussian binomial coefficients.** For a real number u and integers $n \geq 0$ and k, the Gaussian binomial coefficient is

$$\begin{bmatrix} n \\ k \end{bmatrix}_u = \prod_{i=0}^{k-1} \frac{1-u^{n-i}}{1-u^{i+1}} \quad (0 \leq k \leq n), \qquad \begin{bmatrix} n \\ k \end{bmatrix}_u = 0 \quad (k < 0 \text{ or } k > n),$$

interpreted as a polynomial in u. We never use a negative top parameter. Empty products are equal to 1. We recall the following standard facts; see [1, Chapter 3]. First,

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = q^{k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_{q^{-1}}. \tag{2.1}$$

Second, the Pascal recurrence

$$\begin{bmatrix} n \\ k \end{bmatrix}_u = u^k \begin{bmatrix} n-1 \\ k \end{bmatrix}_u + \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_u \qquad (n \geq 1). \tag{2.2}$$

Third, for integers $0 \leq a \leq b \leq n$,

$$\begin{bmatrix} n \\ b \end{bmatrix}_u \begin{bmatrix} b \\ a \end{bmatrix}_u = \begin{bmatrix} n \\ a \end{bmatrix}_u \begin{bmatrix} n-a \\ b-a \end{bmatrix}_u, \tag{2.3}$$

and both sides vanish if $b > n$. Fourth, the finite q-binomial theorem:

$$\sum_{h=0}^{s} (-1)^h u^{\binom{h}{2}} \begin{bmatrix} s \\ h \end{bmatrix}_u x^h = \prod_{i=0}^{s-1} (1 - xu^i). \tag{2.4}$$

Finally, if q is a prime power, then $\begin{bmatrix} n \\ k \end{bmatrix}_q$ is the number of k-dimensional subspaces of $\mathbb{F}_q^n$. For $0 < u < 1$ and $0 \leq k \leq n$ we also write $\begin{bmatrix} n \\ k \end{bmatrix}_u = (u;u)_n / ((u;u)_k (u;u)_{n-k})$.

2.2. **Kazhdan–Lusztig invariants and incidence inversion.** We follow [3, Section 2.2]. To every loopless matroid M are attached polynomials P_M, Z_M, Q_M and Y_M in $\mathbb{Z}[t]$, uniquely determined by the conditions of [3, Theorems 2.1 and 2.2]: $P_M = Q_M = 1$ if $\operatorname{rk} M = 0$; $\deg P_M < \frac{1}{2} \operatorname{rk} M$ and $\deg Q_M < \frac{1}{2} \operatorname{rk} M$ if $\operatorname{rk} M > 0$; and the polynomials

$$Z_M(t) = \sum_{F \in L(M)} t^{\operatorname{rk} F} P_{M/F}(t) \tag{2.5}$$

and $Y_M(t)$, given by (1.1), are palindromic. All four polynomials depend only on the lattice of flats $L(M)$, so they are unchanged under any isomorphism of lattices of flats.

For an interval $[F, G]$ of $L(M)$ let $M_{[F,G]} = (M|G)/F$. In the incidence algebra of $L(M)$ over $\mathbb{Z}[t, t^{-1}]$, let Z and $\widehat{Y}$ be the elements with $Z_{[F,G]} = Z_{M_{[F,G]}}$ and $\widehat{Y}_{[F,G]} = (-1)^{\operatorname{rk} M_{[F,G]}} Y_{M_{[F,G]}}$. Then Z and $\widehat{Y}$ are mutually inverse [3, Section 2.2]. Evaluating the identity $Z\widehat{Y} = 1$ on the interval $[\varnothing, E]$, we obtain, for every loopless matroid M of positive rank,

$$\sum_{F \in L(M)} (-1)^{\operatorname{rk} M - \operatorname{rk} F} Z_{M|F}(t)\, Y_{M/F}(t) = 0. \tag{2.6}$$

This is the only property of Y that we use; we do not need Q or μ explicitly.

2.3. **Projective geometries.** For $m \geq 1$ let $\mathrm{PG}(m-1,q)$ denote the simple matroid on $\mathbb{P}(\mathbb{F}_q^m)$ given by linear independence, and let $\mathrm{PG}(-1,q)$ be the matroid of rank zero. The lattice of flats of $\mathrm{PG}(m-1,q)$ is the lattice of subspaces of $\mathbb{F}_q^m$. Put

$$G_m(t) = \sum_{i=0}^{m} \begin{bmatrix} m \\ i \end{bmatrix}_q t^i.$$

Lemma 2.1. *For $m \geq 0$ we have $P_{\mathrm{PG}(m-1,q)} = 1$ and $Z_{\mathrm{PG}(m-1,q)} = G_m$.*

Proof. We use induction on m; the case $m = 0$ is part of the definition. Let $m \geq 1$. Every contraction of $\mathrm{PG}(m-1,q)$ by a nonempty flat of rank d has the lattice of flats of $\mathrm{PG}(m-d-1,q)$, so by induction its Kazhdan–Lusztig polynomial is 1. By (2.5), $Z_{\mathrm{PG}(m-1,q)} = P_{\mathrm{PG}(m-1,q)} + G_m - 1$. Since G_m is palindromic of degree m, the polynomial $P_{\mathrm{PG}(m-1,q)} - 1$ is palindromic of degree m in the sense $t^m f(t^{-1}) = f(t)$, and its degree is less than $m/2$. Hence it is zero. □

Lemma 2.2 ([9, Proposition 3.8])**.** *For $m \geq 0$,*

$$J_m(t) := Y_{\mathrm{PG}(m-1,q)}(t) = \sum_{i=0}^{m} q^{\binom{i}{2}+\binom{m-i}{2}} \begin{bmatrix} m \\ i \end{bmatrix}_q t^i.$$

Equivalently, by (2.1)*, $J_m(t) = q^{\binom{m}{2}} \sum_{i=0}^{m} \begin{bmatrix} m \\ i \end{bmatrix}_{q^{-1}} t^i$.*

Proof. For $m \geq 1$, the matroid $\mathrm{PG}(m-1,q)$ is the q-niform matroid $U_{m,m}(q)$ of [9], and [9, Proposition 3.8] with $n = k = m$ gives the coefficient of t^i for $0 \leq i \leq \lfloor m/2 \rfloor$ and of t^{m-i} for $0 \leq i \leq \lfloor (m-1)/2 \rfloor$; since $\begin{bmatrix} m \\ i \end{bmatrix}_q = \begin{bmatrix} m \\ m-i \end{bmatrix}_q$, the two halves combine to the displayed formula, the middle coefficient for even m being counted once. For $m = 0$ we have $Y = 1 = J_0$ by definition. The second formula follows from $i(m-i) + \binom{i}{2} + \binom{m-i}{2} = \binom{m}{2}$. □

2.4. **Bose–Burton matroids.** Throughout, q is a prime power, V is a vector space of dimension r over $\mathbb{F}_q$, $W \subseteq V$ is a subspace of dimension s, and $B = B_{r,s}(q)$ is the matroid described in the introduction, with ground set $E = \mathbb{P}(V) \setminus \mathbb{P}(W)$. Its rank function is $\mathrm{rk}(A) = \dim\langle A \rangle$, where $\langle A \rangle$ is the subspace spanned by the points of A. It is convenient to allow $s = 0$ and to write $B_{d,0}(q) = \mathrm{PG}(d-1,q)$. For a subspace $U \subseteq V$ put $E_U = \mathbb{P}(U) \setminus \mathbb{P}(W)$.

Lemma 2.3. *Let $r \geq 1$ and $0 \leq s \leq r-1$.*

(a) *For every subspace $U \not\subseteq W$ we have $\langle E_U \rangle = U$. The map $U \mapsto E_U$ is an inclusion-preserving bijection from the set of subspaces $U \not\subseteq W$ onto the set of nonempty flats of B, and $\mathrm{rk}\, E_U = \dim U$. In particular B is a simple matroid of rank r.*

(b) *If $\dim U = d$ and $\dim(U \cap W) = a$, then $0 \leq a \leq d-1$ and $B|E_U \cong B_{d,a}(q)$.*

(c) *For a nonempty flat E_U with $\dim U = d$, the lattice of flats of B/E_U is isomorphic to the lattice of subspaces of V/U, that is, to the lattice of flats of $\mathrm{PG}(r-d-1,q)$. Consequently $Y_{B/E_U} = J_{r-d}$.*

(d) *For integers $1 \leq d \leq r$ and $0 \leq a \leq d-1$, the number of subspaces $U \subseteq V$ with $\dim U = d$ and $\dim(U \cap W) = a$ is*

$$N(d,a) = q^{(s-a)(d-a)} \begin{bmatrix} s \\ a \end{bmatrix}_q \begin{bmatrix} r-s \\ d-a \end{bmatrix}_q. \tag{2.7}$$

It is nonzero exactly when $\max(0, d-r+s) \leq a \leq \min(s, d-1)$.

Proof. (a) Let $U \not\subseteq W$ and choose $u \in U \setminus W$. Every vector of $U \setminus W$ represents a point of E_U. If $v \in U \cap W$, then u and $u+v$ both lie in $U \setminus W$, so $v = (u+v) - u \in \langle E_U \rangle$. Hence $\langle E_U \rangle = U$. The closure of a set $A \subseteq E$ in B is $\mathbb{P}(\langle A \rangle) \cap E = E_{\langle A \rangle}$, so the flats of B are the sets E_U; this set is empty if $U \subseteq W$, and if $U \not\subseteq W$ it determines $U = \langle E_U \rangle$. The remaining assertions follow, taking $U = V$ for the rank.

(b) We have $a \le d-1$ because $U \not\subseteq W$, and $E_U = \mathbb{P}(U) \setminus \mathbb{P}(U \cap W)$ with the rank function of U.

(c) The flats of B/E_U are the sets $F \setminus E_U$ with $F \supseteq E_U$ a flat of B, and by (a) these correspond, preserving inclusion, to the subspaces U' with $U \subseteq U' \subseteq V$; each such U' satisfies $U' \not\subseteq W$. The interval $[U, V]$ of the subspace lattice is isomorphic to the subspace lattice of V/U. Since Y depends only on the lattice of flats, Lemma 2.2 gives $Y_{B/E_U} = J_{r-d}$; for $d = r$ the contraction has rank zero and $Y = 1 = J_0$.

(d) The subspace $A = U \cap W$ can be chosen in $\begin{bmatrix} s \\ a \end{bmatrix}_q$ ways. Given A, the image $(U+W)/W$ is a $(d-a)$-dimensional subspace of V/W, chosen in $\begin{bmatrix} r-s \\ d-a \end{bmatrix}_q$ ways. Given A and this image, with preimage $X \subseteq V$ of dimension $s+d-a$, the subspace U/A is a complement of W/A in X/A; there are $q^{(s-a)(d-a)}$ such complements. The last assertion is clear. □

The next lemma is due to Cheng and Liu [5, Remark 2.7], who state it for all $0 \le a < d$ (only their further simplification $P = G_a$, $Z = G_d$ requires $a < d/2$). We include the short argument, which uses only Lemmas 2.1 and 2.3.

Lemma 2.4 ([5, Remark 2.7])**.** *Let $d \ge 1$ and $0 \le a \le d-1$. Then*

$$P_{B_{d,a}(q)}(t) = \sum_{0 \le i < d/2} \left(\begin{bmatrix} a \\ i \end{bmatrix}_q - \begin{bmatrix} a \\ d-i \end{bmatrix}_q \right) t^i, \qquad Z_{B_{d,a}(q)} = P_{B_{d,a}(q)} + G_d - G_a.$$

Consequently, for $0 \le i \le d$,

$$[t^i]\, Z_{B_{d,a}(q)}(t) = \begin{bmatrix} d \\ i \end{bmatrix}_q - \begin{bmatrix} a \\ \max(i, d-i) \end{bmatrix}_q. \tag{2.8}$$

Proof. By Lemma 2.3(c) and Lemma 2.1, every contraction of $B_{d,a}(q)$ by a nonempty flat has Kazhdan–Lusztig polynomial 1. By Lemma 2.3(a), the number of flats of rank $i \ge 1$ is $\begin{bmatrix} d \\ i \end{bmatrix}_q - \begin{bmatrix} a \\ i \end{bmatrix}_q$. Hence (2.5) gives $Z_{B_{d,a}(q)} = P_{B_{d,a}(q)} + G_d - G_a$. Write $P_{B_{d,a}(q)} = \sum_i p_i t^i$, with $p_i = 0$ for $i \ge d/2$. For $i < d/2$, palindromicity of $Z_{B_{d,a}(q)}$ compares the coefficient $p_i + \begin{bmatrix} d \\ i \end{bmatrix}_q - \begin{bmatrix} a \\ i \end{bmatrix}_q$ of t^i with the coefficient $\begin{bmatrix} d \\ d-i \end{bmatrix}_q - \begin{bmatrix} a \\ d-i \end{bmatrix}_q$ of t^{d-i}, and gives $p_i = \begin{bmatrix} a \\ i \end{bmatrix}_q - \begin{bmatrix} a \\ d-i \end{bmatrix}_q$. Formula (2.8) follows by treating $i < d/2$ and $i \ge d/2$ separately. □

3. A signed formula for the inverse Z-polynomial

In this section $r \ge 2$ and $1 \le s \le r-1$. For integers $1 \le d \le r$ and $0 \le i \le d$ we write

$$h(d,i) = \max(i, d-i), \qquad \ell(d,i) = \min(i, d-i).$$

Lemma 3.1. *Let $1 \le d \le r$, $0 \le i \le d$ and $h = h(d,i)$. Then*

$$\sum_a N(d,a)\, [t^i] Z_{B_{d,a}(q)}(t) = \begin{bmatrix} r \\ d \end{bmatrix}_q \begin{bmatrix} d \\ i \end{bmatrix}_q - \begin{bmatrix} s \\ h \end{bmatrix}_q \begin{bmatrix} r-h \\ d-h \end{bmatrix}_q,$$

where the sum is over $\max(0, d-r+s) \le a \le \min(s, d-1)$.

Proof. For $0 \le h' \le s$, counting pairs (I, U) of subspaces with $I \subseteq W$, $\dim I = h'$, $I \subseteq U$, $\dim U = d$ and $U \not\subseteq W$ in two ways gives

$$\sum_a N(d,a) \begin{bmatrix} a \\ h' \end{bmatrix}_q = \begin{bmatrix} s \\ h' \end{bmatrix}_q \left(\begin{bmatrix} r-h' \\ d-h' \end{bmatrix}_q - \begin{bmatrix} s-h' \\ d-h' \end{bmatrix}_q \right);$$

the second term removes the subspaces $U \subseteq W$. In particular $\sum_a N(d,a) = \begin{bmatrix} r \\ d \end{bmatrix}_q - \begin{bmatrix} s \\ d \end{bmatrix}_q$. If $h \le s$, then (2.8) and these identities give

$$\sum_a N(d,a)[t^i] Z_{B_{d,a}(q)} = \begin{bmatrix} d \\ i \end{bmatrix}_q \left(\begin{bmatrix} r \\ d \end{bmatrix}_q - \begin{bmatrix} s \\ d \end{bmatrix}_q \right) - \begin{bmatrix} s \\ h \end{bmatrix}_q \begin{bmatrix} r-h \\ d-h \end{bmatrix}_q + \begin{bmatrix} s \\ h \end{bmatrix}_q \begin{bmatrix} s-h \\ d-h \end{bmatrix}_q .$$

By (2.3) and $\begin{bmatrix} d \\ h \end{bmatrix}_q = \begin{bmatrix} d \\ i \end{bmatrix}_q$, the last term equals $\begin{bmatrix} s \\ d \end{bmatrix}_q \begin{bmatrix} d \\ i \end{bmatrix}_q$, which cancels. If $h > s$, then $\begin{bmatrix} a \\ h \end{bmatrix}_q = 0$ for all $a \le s$, and also $\begin{bmatrix} s \\ d \end{bmatrix}_q = 0$ because $d \ge h > s$; the claim holds with $\begin{bmatrix} s \\ h \end{bmatrix}_q = 0$. □

For integers $0 \le \ell \le h$ put $E_{h,\ell}(t) = t^\ell + t^h$ if $\ell < h$ and $E_{h,h}(t) = t^h$.

Proposition 3.2. *For $r \ge 2$ and $1 \le s \le r-1$,*

$$Y_{B_{r,s}(q)}(t) = J_r(t) + \sum_{h=1}^{s} \sum_{\ell=0}^{\min(h,r-h)} (-1)^{h+\ell} \begin{bmatrix} s \\ h \end{bmatrix}_q \begin{bmatrix} r-h \\ \ell \end{bmatrix}_q E_{h,\ell}(t)\, J_{r-h-\ell}(t). \tag{3.1}$$

Proof. Apply (2.6) to $M = B = B_{r,s}(q)$. The flat $\varnothing$ contributes $(-1)^r Y_B$. By Lemma 2.3, the remaining flats are the sets E_U, and a flat with $\dim U = d$, $\dim(U \cap W) = a$ contributes $(-1)^{r-d} Z_{B_{d,a}(q)} J_{r-d}$. Multiplying by $(-1)^r$ we obtain

$$Y_B = -\sum_{d=1}^{r} (-1)^d \sum_a N(d,a)\, Z_{B_{d,a}(q)}\, J_{r-d}. \tag{3.2}$$

All flats are included, among them E itself ($d = r$, $a = s$), whose contraction has rank zero. The same argument for $\mathrm{PG}(r-1,q)$, using Lemma 2.1, gives

$$J_r = -\sum_{d=1}^{r} (-1)^d \begin{bmatrix} r \\ d \end{bmatrix}_q G_d\, J_{r-d}.$$

Subtracting and applying Lemma 3.1 coefficientwise, we get

$$Y_B - J_r = \sum_{d=1}^{r} (-1)^d \sum_{i=0}^{d} \begin{bmatrix} s \\ h(d,i) \end{bmatrix}_q \begin{bmatrix} r-h(d,i) \\ d-h(d,i) \end{bmatrix}_q t^i J_{r-d}.$$

The map $(d,i) \mapsto (h,\ell) = (h(d,i), \ell(d,i))$ sends the pairs with $1 \le d \le r$, $0 \le i \le d$ onto the pairs with $h \ge 1$, $0 \le \ell \le \min(h, r-h)$; it is two-to-one over pairs with $\ell < h$, the preimages having $i = \ell$ and $i = h$, and one-to-one over pairs with $\ell = h$. Since $d = h + \ell$, $d - h = \ell$ and $(-1)^d = (-1)^{h+\ell}$, and since terms with $h > s$ vanish, we obtain (3.1). □

Example 3.3. For $s = 1$ only $h = 1$ occurs, and (3.1) reads

$$Y_{B_{r,1}(q)} = J_r - (1+t) J_{r-1} + \begin{bmatrix} r-1 \\ 1 \end{bmatrix}_q t\, J_{r-2} \qquad (r \ge 2).$$

Example 3.4. For $r = 2$ we have $J_2 = q + (q+1)t + qt^2$, $J_1 = 1 + t$ and $J_0 = 1$, so Example 3.3 gives

$$Y_{U_{2,q}} = Y_{B_{2,1}(q)} = (q-1) + qt + (q-1)t^2,$$

in agreement with the formula for uniform matroids [10, Theorem 1.1]. For $q = 2$ this is $(1+t)^2$, the inverse Z-polynomial of the Boolean matroid of rank two.

4. The normalized coefficients

Throughout this section $r \geq 2$, $1 \leq s \leq r-1$, and

$$z = q^{-1} \in (0, \tfrac{1}{2}], \qquad c = r - s \geq 1, \qquad g_k = \begin{bmatrix} r \\ k \end{bmatrix}_z.$$

Write $y_k = [t^k] Y_{B_{r,s}(q)}$ and $a_k = q^{-\binom{r}{2}} y_k$. By Lemma 2.2 and (2.1), $\pi_k = q^{\binom{r}{2}} g_k$, so $a_k = g_k F_k$ with $F_k = y_k/\pi_k$ as in Theorem 1.2. Put $E_{0,0} = 1$, so that the term J_r in (3.1) is the term $h = \ell = 0$ of the double sum.

Lemma 4.1. *For $0 \leq k \leq r$,*

$$a_k = \sum_{h=0}^{s} \sum_{\ell=0}^{\min(h, r-h)} (-1)^{h+\ell} z^{hc + \binom{h}{2} + \binom{\ell}{2}} \begin{bmatrix} s \\ h \end{bmatrix}_z \begin{bmatrix} r-h \\ \ell \end{bmatrix}_z \times \left(\begin{bmatrix} r-h-\ell \\ k-\ell \end{bmatrix}_z + [\ell < h] \begin{bmatrix} r-h-\ell \\ k-h \end{bmatrix}_z \right),$$

where $[\ell < h]$ is 1 if $\ell < h$ and 0 otherwise.

Proof. By (2.1) and Lemma 2.2, $J_m = q^{\binom{m}{2}} \sum_i \begin{bmatrix} m \\ i \end{bmatrix}_z t^i$, and

$$\begin{bmatrix} s \\ h \end{bmatrix}_q \begin{bmatrix} r-h \\ \ell \end{bmatrix}_q q^{\binom{r-h-\ell}{2} - \binom{r}{2}} = z^{hc + \binom{h}{2} + \binom{\ell}{2}} \begin{bmatrix} s \\ h \end{bmatrix}_z \begin{bmatrix} r-h \\ \ell \end{bmatrix}_z.$$

Indeed, the exponent of q on the left, after applying (2.1), is $h(s-h) + \ell(r-h-\ell) + \binom{r-h-\ell}{2} - \binom{r}{2}$; expanding $\binom{r}{2} = \binom{r-h-\ell}{2} + \binom{h}{2} + \binom{\ell}{2} + h\ell + (h+\ell)(r-h-\ell)$ shows that it equals $-hc - \binom{h}{2} - \binom{\ell}{2}$. The coefficient of t^k in $E_{h,\ell}(t) \sum_i \begin{bmatrix} m \\ i \end{bmatrix}_z t^i$ with $m = r-h-\ell$ is the bracket in the statement. □

Lemma 4.2. *Let $n \geq 1$ and $0 \leq b \leq n$. Then*

$$\sum_{\ell=0}^{b} (-1)^\ell z^{\binom{\ell}{2}} \begin{bmatrix} n \\ \ell \end{bmatrix}_z = (-1)^b z^{\binom{b+1}{2}} \begin{bmatrix} n-1 \\ b \end{bmatrix}_z.$$

In particular the sum vanishes for $b = n$.

Proof. Put $A_\ell = (-1)^\ell z^{\binom{\ell+1}{2}} \begin{bmatrix} n-1 \\ \ell \end{bmatrix}_z$ for $\ell \geq -1$, so $A_{-1} = 0$. By (2.2),

$$(-1)^\ell z^{\binom{\ell}{2}} \begin{bmatrix} n \\ \ell \end{bmatrix}_z = (-1)^\ell z^{\binom{\ell+1}{2}} \begin{bmatrix} n-1 \\ \ell \end{bmatrix}_z + (-1)^\ell z^{\binom{\ell}{2}} \begin{bmatrix} n-1 \\ \ell-1 \end{bmatrix}_z = A_\ell - A_{\ell-1}.$$

The sum telescopes to A_b, and $A_n = 0$. □

Proposition 4.3. $F_0 = F_r = \prod_{i=c}^{r-1} (1 - z^i)$.

Proof. For $k = 0$, the first term of the bracket in Lemma 4.1 is nonzero only if $\ell = 0$, when it equals 1, and the second term vanishes because it occurs only for $h \geq 1$, where $k - h < 0$. Hence, by (2.4) with $x = z^c$,

$$F_0 = a_0 = \sum_{h=0}^{s} (-1)^h z^{\binom{h}{2}} \begin{bmatrix} s \\ h \end{bmatrix}_z z^{ch} = \prod_{i=0}^{s-1} (1 - z^{c+i}) = \prod_{i=c}^{r-1} (1 - z^i).$$

For each pair (h, ℓ), the polynomial $H(t) = E_{h,\ell}(t) J_{r-h-\ell}(t)$ satisfies $t^r H(t^{-1}) = H(t)$, because $t^{h+\ell} E_{h,\ell}(t^{-1}) = E_{h,\ell}(t)$ and $t^m J_m(t^{-1}) = J_m(t)$. Thus the coefficients of H are symmetric about $r/2$ (its degree is $r - \ell$, so its coefficients of $t^0, \ldots, t^{\ell-1}$ vanish together with those of $t^{r-\ell+1}, \ldots, t^r$). Hence $y_k = y_{r-k}$ by (3.1) (this is also [10, Lemma 2.2]). Since $g_k = g_{r-k}$, we get $F_r = F_0$. □

For $1 \leq k \leq r - 1$ and $h \geq 0$ define

$$T_h(k) = z^{h(c+h)} \begin{bmatrix} s \\ h \end{bmatrix}_z \begin{bmatrix} r-h \\ k \end{bmatrix}_z \begin{bmatrix} k-1 \\ h \end{bmatrix}_z,$$

$$U_h(k) = z^{h(c+h-1)} \begin{bmatrix} s \\ h \end{bmatrix}_z \begin{bmatrix} r-h \\ k-h \end{bmatrix}_z \begin{bmatrix} r-k-1 \\ h-1 \end{bmatrix}_z,$$

the latter for $h \geq 1$. All top parameters are nonnegative for the indices used below.

Proposition 4.4. *For $1 \leq k \leq r - 1$,*

$$a_k = \sum_{h=0}^{\min(s,k-1,r-k)} T_h(k) - \sum_{h=1}^{\min(s,k,r-k)} U_h(k).$$

Proof. Split the bracket in Lemma 4.1 into its two terms.

First terms. By (2.3), $\begin{bmatrix} r-h \\ \ell \end{bmatrix}_z \begin{bmatrix} r-h-\ell \\ k-\ell \end{bmatrix}_z = \begin{bmatrix} r-h \\ k \end{bmatrix}_z \begin{bmatrix} k \\ \ell \end{bmatrix}_z$. This vanishes unless $h \leq r - k$, and then $\min(h, r - h) \geq \min(h, k)$, while $\begin{bmatrix} k \\ \ell \end{bmatrix}_z = 0$ for $\ell > k$. So the contribution of h is

$$(-1)^h z^{hc+\binom{h}{2}} \begin{bmatrix} s \\ h \end{bmatrix}_z \begin{bmatrix} r-h \\ k \end{bmatrix}_z \sum_{\ell=0}^{\min(h,k)} (-1)^\ell z^{\binom{\ell}{2}} \begin{bmatrix} k \\ \ell \end{bmatrix}_z.$$

If $h \geq k$ the inner sum is complete and vanishes by Lemma 4.2. If $h \leq k - 1$, Lemma 4.2 with $n = k$ and $b = h$ turns the contribution into $T_h(k)$, since $\binom{h}{2} + \binom{h+1}{2} = h^2$. Thus the first terms contribute $\sum_{h \leq \min(s,k-1,r-k)} T_h(k)$.

Second terms. These occur only for $h \geq 1$ and $\ell \leq \min(h - 1, r - h)$. By (2.3), $\begin{bmatrix} r-h \\ \ell \end{bmatrix}_z \begin{bmatrix} r-h-\ell \\ k-h \end{bmatrix}_z = \begin{bmatrix} r-h \\ k-h \end{bmatrix}_z \begin{bmatrix} r-k \\ \ell \end{bmatrix}_z$. This vanishes unless $h \leq k$, and then $r-h \geq r-k$, while $\begin{bmatrix} r-k \\ \ell \end{bmatrix}_z = 0$ for $\ell > r - k$. So the contribution of h is

$$(-1)^h z^{hc+\binom{h}{2}} \begin{bmatrix} s \\ h \end{bmatrix}_z \begin{bmatrix} r-h \\ k-h \end{bmatrix}_z \sum_{\ell=0}^{\min(h-1,r-k)} (-1)^\ell z^{\binom{\ell}{2}} \begin{bmatrix} r-k \\ \ell \end{bmatrix}_z.$$

If $h-1 \geq r-k$ the inner sum is complete and vanishes. If $h \leq r-k$, Lemma 4.2 with $n = r-k$ and $b = h-1$ turns the contribution into $-U_h(k)$, since $2\binom{h}{2} = h(h-1)$. □

Note that $T_0(k) = g_k$. We now prove Theorem 1.2(ii). Recall $R_j(k)$ and w_j from (1.3) and (1.4), and for $1 \leq h \leq s$ and $1 \leq k \leq r - 1$ put

$$D_h = z^{h(c+h-1)} \begin{bmatrix} s \\ h \end{bmatrix}_z \frac{(z;z)_{r-h}}{(z;z)_r (z;z)_h}, \qquad N_h(k) = 1 - 2z^h - z^r + z^h\big(z^k + z^{r-k}\big).$$

Theorem 4.5. *For $1 \le k \le r-1$,*

$$F_k = \sum_{j=0}^{s} w_j \, R_j(k). \tag{4.1}$$

Moreover $w_j > 0$ for $0 \le j \le s$, $w_0 = (1-z^c)/(1-z^r)$, and $R_j(k) > 0$ if $j < \min(k, r-k)$, while $R_j(k) = 0$ if $j \ge \min(k, r-k)$.

Proof. Step 1. We show that

$$F_k = 1 - \sum_{h=1}^{s} D_h \, R_{h-1}(k) \, N_h(k). \tag{4.2}$$

We write every Gaussian coefficient as a quotient of the symbols $(z;z)_m$; in particular $g_k = (z;z)_r/((z;z)_k\,(z;z)_{r-k})$. A direct cancellation then gives, whenever the respective term occurs in Proposition 4.4,

$$\begin{aligned} U_h(k)/g_k &= D_h \, R_{h-1}(k) \, (1-z^h)(1-z^k), \\ T_h(k)/g_k &= D_h \, R_{h-1}(k) \, z^h (1-z^{k-h})(1-z^{r-k}). \end{aligned}$$

For instance,

$$\frac{U_h(k)}{g_k} = z^{h(c+h-1)} \begin{bmatrix} s \\ h \end{bmatrix}_z \frac{(z;z)_{r-h}}{(z;z)_r \, (z;z)_{h-1}} \cdot \frac{(z;z)_k}{(z;z)_{k-h}} \cdot \frac{(z;z)_{r-k-1}}{(z;z)_{r-k-h}},$$

where $(z;z)_{h-1}^{-1} = (1-z^h)(z;z)_h^{-1}$ and the product of the last two quotients is $(1-z^k)R_{h-1}(k)$; the computation for $T_h(k)$ is the same. A short expansion shows

$$z^h(1-z^{k-h})(1-z^{r-k}) - (1-z^h)(1-z^k) = -N_h(k).$$

Now let $1 \le h \le s$. If $h \le \min(k-1, r-k)$, both $T_h(k)$ and $U_h(k)$ occur, and $(U_h - T_h)/g_k = D_h R_{h-1} N_h$. If $h = k \le r-k$, only $U_h(k)$ occurs; the displayed formula for $T_h(k)/g_k$ vanishes because of the factor $1 - z^{k-h}$, so again $U_h/g_k = D_h R_{h-1} N_h$. If $h > \min(k, r-k)$, neither term occurs, and $R_{h-1}(k) = 0$ because it contains the factor $1 - z^0$ (from $i = k$ or $i = r-k$, both at most $h-1$). Since $T_0(k)/g_k = 1$, Proposition 4.4 gives (4.2).

Step 2. Since $R_h(k) = R_{h-1}(k)(1-z^{k-h})(1-z^{r-k-h})$, expanding the right-hand side shows

$$R_{h-1}(k)N_h(k) = (1-z^h)^2 R_{h-1}(k) - z^{2h} R_h(k). \tag{4.3}$$

Substituting (4.3) in (4.2) and collecting the coefficient of each $R_j(k)$, we obtain

$$F_k = \big(1 - D_1(1-z)^2\big) + \sum_{j=1}^{s-1} \big(D_j z^{2j} - D_{j+1}(1-z^{j+1})^2\big) R_j(k) + D_s z^{2s} R_s(k).$$

For $1 \le j \le s-1$,

$$\frac{D_{j+1}}{D_j} = \frac{z^{c+2j}(1-z^{s-j})}{(1-z^{j+1})^2(1-z^{r-j})},$$

so, using $c + s = r$,

$$D_j z^{2j} - D_{j+1}(1-z^{j+1})^2 = D_j z^{2j} \, \frac{1 - z^{r-j} - z^c + z^{r-j}}{1-z^{r-j}} = D_j z^{2j} \frac{1-z^c}{1-z^{r-j}}.$$

The same expression is valid for $j = s$, since then $(1-z^c)/(1-z^{r-j}) = 1$. Moreover $D_1(1-z)^2 = z^c(1-z^s)/(1-z^r)$, so $1 - D_1(1-z)^2 = (1-z^c)/(1-z^r)$. Finally,

$$D_j z^{2j} \frac{1-z^c}{1-z^{r-j}} = (1-z^c) z^{j(c+j+1)} \begin{bmatrix} s \\ j \end{bmatrix}_z \frac{(z;z)_{r-j-1}}{(z;z)_r (z;z)_j} = w_j,$$

and $w_0 = (1-z^c)/(1-z^r)$. This proves (4.1).

Step 3. Since $0 < z < 1$, $c \geq 1$ and $r-j-1 \geq c-1 \geq 0$ for $j \leq s$, every factor of w_j is positive. If $j < \min(k, r-k)$, all exponents $k-i$ and $r-k-i$ with $1 \leq i \leq j$ are positive, so $R_j(k) > 0$. If $j \geq \min(k, r-k)$, the product $R_j(k)$ contains the factor with $i = k$ or $i = r-k$, which is $1 - z^0 = 0$. □

Corollary 4.6. *The numbers $F_0, \ldots, F_r$ are positive and $F_k = F_{r-k}$. Moreover $F_1 = F_{r-1} = w_0 = (1-z^c)/(1-z^r)$.*

Proof. Positivity follows from Proposition 4.3 and Theorem 4.5, since $w_0 R_0(k) = w_0 > 0$. Symmetry is part of Proposition 4.3 for the endpoints and is evident from (4.1) for $1 \leq k \leq r-1$, because $R_j(k) = R_j(r-k)$. For $k \in \{1, r-1\}$ we have $\min(k, r-k) = 1$, so only $j = 0$ contributes. □

5. Concavity of the normalized coefficients

We keep the notation of Section 4. Here the hypothesis that $q \geq 2$ enters through $z \leq \frac{1}{2}$, that is, $z \leq 1 - z$.

Lemma 5.1. *Let $j \geq 1$ and, for integers $k \geq j$, put $f_j(k) = \prod_{i=1}^{j}(1-z^{k-i})$. Then $f_j(j) = 0$, and for $k \geq j$*

$$f_j(k+1) - f_j(k) = z^{k-j}(1-z^j) \prod_{i=1}^{j-1}(1-z^{k-i}) > 0. \tag{5.1}$$

Moreover the increments are nonincreasing: $f_j(k+1) - f_j(k) \leq f_j(k) - f_j(k-1)$ for $k \geq j+1$.

Proof. We have

$$f_j(k+1) - f_j(k) = \prod_{i=1}^{j-1}(1-z^{k-i}) \cdot \big((1-z^k) - (1-z^{k-j})\big),$$

which is (5.1); all factors are positive for $k \geq j$. For $k \geq j+1$, both k and $k-1$ are at least j, and (5.1) gives the ratio of consecutive increments

$$\frac{f_j(k+1) - f_j(k)}{f_j(k) - f_j(k-1)} = \frac{z(1-z^{k-1})}{1-z^{k-j}} \leq \frac{z}{1-z} \leq 1,$$

where we used $1 - z^{k-1} \leq 1$ and $1 - z^{k-j} \geq 1 - z$, as $k - j \geq 1$. □

Lemma 5.2. *Let $j \geq 1$ and let k be an integer with $j+1 \leq k \leq r-j-1$. Then*

$$R_j(k+1) + R_j(k-1) - 2R_j(k) \leq 0.$$

Proof. We have $R_j(m) = f_j(m) f_j(r-m)$. Put $a = f_j(k)$, $b = f_j(r-k)$ and

$$u = f_j(k+1) - f_j(k), \qquad v = f_j(k) - f_j(k-1),$$
$$u' = f_j(r-k+1) - f_j(r-k), \qquad v' = f_j(r-k) - f_j(r-k-1).$$

Then $R_j(k) = ab$, $R_j(k+1) = (a+u)(b-v')$ and $R_j(k-1) = (a-v)(b+u')$. The hypotheses give $k-1 \geq j$ and $r-k-1 \geq j$, so Lemma 5.1 applies at k and at $r-k$, and $a, b \geq 0$, $0 \leq u \leq v$, $0 \leq u' \leq v'$. Then

$$R_j(k+1) + R_j(k-1) - 2R_j(k) = b(u-v) + a(u'-v') - uv' - vu' \leq 0. \qquad \square$$

Remark 5.3. Lemma 5.2 does not extend to $k = j$. Let $2 \leq j \leq s$ and $r \geq 2j+2$, so that the indices $j-1, j, j+1$ lie in $[1, r-1]$. Then $R_j(j-1) = R_j(j) = 0 < R_j(j+1)$, so the second difference of R_j at $k = j$ is positive; since $R_j(k) = R_j(r-k)$, the same holds at the reflected index $k = r-j$. These are the only internal indices, with both neighbours internal, at which a term of (4.1) has positive second difference: for $j+1 \leq k \leq r-j-1$ Lemma 5.2 applies, and for $k \leq j-1$ or $k \geq r-j+1$ the product R_j vanishes at $k-1$, k and $k+1$. Proposition 5.4 compensates the term at $k = j$ by the term $j-1$, and the reflected index is handled by symmetry. For $j = 1$ the index $k = 1$ has the neighbour 0, where (4.1) is not used; the indices 1 and $r-1$ are treated in Proposition 5.5.

Proposition 5.4. *For $2 \leq k \leq r-2$,*

$$2F_k \geq F_{k-1} + F_{k+1}.$$

Proof. By Corollary 4.6, $F_m = F_{r-m}$ for all m, so the inequality at k is equivalent to the inequality at $r-k$. We may therefore assume $2 \leq k \leq r/2$. All three indices $k-1, k, k+1$ lie in $[1, r-1]$, so (4.1) applies to each of them, and

$$F_{k+1} + F_{k-1} - 2F_k = \sum_{j=0}^{s} w_j\, \delta_j, \qquad \delta_j = R_j(k+1) + R_j(k-1) - 2R_j(k).$$

We have $\delta_0 = 0$. For $1 \leq j \leq k-1$ we have $j+1 \leq k$ and $r-k \geq k \geq j+1$, so $\delta_j \leq 0$ by Lemma 5.2. For $j \geq k+1$, R_j vanishes at $k-1$, k and $k+1$ by Theorem 4.5, so $\delta_j = 0$. It remains to consider $j = k$, which occurs only if $k \leq s$. Since $R_k(k-1) = R_k(k) = 0$, we have $\delta_k = R_k(k+1)$, which is positive if $r \geq 2k+2$ and zero otherwise. Thus, if $k > s$ or $r \leq 2k+1$, every δ_j is nonpositive and we are done. (This covers the central indices $r = 2k$ and $r = 2k+1$.)

Assume now $k \leq s$ and $r \geq 2k+2$. It suffices to show $w_k \delta_k \leq -w_{k-1}\delta_{k-1}$. Since $k-1 \geq 1$, we have $R_{k-1}(k-1) = 0$ and $R_{k-1}(k) > 0$, and a direct computation from the definition gives

$$\frac{R_{k-1}(k+1)}{R_{k-1}(k)} = \frac{(1-z^k)(1-z^{r-2k})}{(1-z)(1-z^{r-k-1})} \leq \frac{1-z^k}{1-z}, \tag{5.2}$$

because $1 \leq r-2k \leq r-k-1$. Hence

$$-\frac{\delta_{k-1}}{R_{k-1}(k)} = 2 - \frac{R_{k-1}(k+1)}{R_{k-1}(k)} \geq \frac{1-2z+z^k}{1-z} \geq \frac{z^k}{1-z}.$$

On the other hand, from (1.4),

$$\frac{w_k}{w_{k-1}} = \frac{z^{c+2k}(1-z^{s-k+1})}{(1-z^k)^2(1-z^{r-k})},$$

and a direct computation gives

$$\frac{R_k(k+1)}{R_{k-1}(k)} = \frac{(1-z^k)(1-z^{r-2k-1})(1-z^{r-2k})}{1-z^{r-k-1}},$$

so that

$$\frac{w_k\,\delta_k}{w_{k-1}R_{k-1}(k)} = \frac{z^{c+2k}(1-z^{s-k+1})(1-z^{r-2k-1})(1-z^{r-2k})}{(1-z^k)(1-z^{r-k})(1-z^{r-k-1})} \le \frac{z^{c+2k}}{(1-z)^3}.$$

Here the three factors in the numerator lie in $(0,1)$ because $s-k+1 \ge 1$ and $r-2k-1 \ge 1$, and the three factors in the denominator are at least $1-z$ because their exponents are positive. Finally, $c \ge 1$, $k \ge 2$ and $z \le \frac{1}{2}$ give $z^{c+k} \le z^3 \le z^2 \le (1-z)^2$, hence

$$\frac{z^{c+2k}}{(1-z)^3} \le \frac{z^k}{1-z} \le -\frac{\delta_{k-1}}{R_{k-1}(k)}.$$

Multiplying by $w_{k-1}R_{k-1}(k) > 0$ gives $w_k\delta_k \le -w_{k-1}\delta_{k-1}$, and therefore $\sum_j w_j\delta_j \le 0$, as claimed. □

Proposition 5.5. *$2F_1 > F_0 + F_2$ and $2F_{r-1} > F_{r-2} + F_r$.*

Proof. By symmetry it suffices to prove the first inequality. By Proposition 4.3 and Corollary 4.6,

$$\frac{F_0}{F_1} = \prod_{i=c}^{r-1}(1-z^i)\cdot\frac{1-z^r}{1-z^c} = \prod_{i=c+1}^{r}(1-z^i),$$

a product with at least one factor, since $c+1 \le r$. Each factor lies in $(0,1)$, so retaining only the first factor gives

$$\frac{F_1 - F_0}{F_1} \ge z^{c+1} > 0. \tag{5.3}$$

If $r = 2$, then $F_2 = F_0$ and the claim is $F_1 > F_0$. If $r = 3$, then $F_2 = F_{r-1} = F_1$ and the claim is again $F_1 > F_0$. Let $r \ge 4$. Then $\min(2, r-2) = 2$, so only $j = 0, 1$ contribute to F_2 in (4.1), and $R_1(2) = (1-z)(1-z^{r-3})$. Using $w_1/w_0 = z^{c+2}(1-z^s)/((1-z)^2(1-z^{r-1}))$, we obtain

$$\frac{F_2 - F_1}{F_1} = \frac{w_1R_1(2)}{w_0} = \frac{z^{c+2}(1-z^s)(1-z^{r-3})}{(1-z)(1-z^{r-1})} < \frac{z^{c+2}}{1-z} \le z^{c+1},$$

where the strict inequality uses $1-z^s < 1$ and $1-z^{r-3} < 1-z^{r-1}$, and the last uses $z \le 1-z$. Together with (5.3), $F_2 - F_1 < F_1 - F_0$. □

Proof of Theorem 1.2. Part (i) is Proposition 4.3, and part (ii) is Theorem 4.5. In part (iii), positivity and symmetry are Corollary 4.6. Concavity at $k = 1$ and $k = r-1$, with strict inequality, is Proposition 5.5, and concavity at $2 \le k \le r-2$ is Proposition 5.4. □

6. Proof of the main theorem and examples

Proof of Theorem 1.1. By Theorem 1.2, $y_k = \pi_k F_k$ with $F_k > 0$, $F_k = F_{r-k}$ and $\pi_k = \pi_{r-k} > 0$. Hence $y_k > 0$ and $y_k = y_{r-k}$. Concavity and positivity of F give, for $1 \le k \le r-1$,

$$F_k \ge \tfrac{1}{2}(F_{k-1} + F_{k+1}) \ge \sqrt{F_{k-1}F_{k+1}},$$

so $F_k^2 \ge F_{k-1}F_{k+1}$. On the other hand $\pi_k = q^{\binom{r}{2}}g_k$ with $g_k = \begin{bmatrix} r \\ k \end{bmatrix}_z$, and for $1 \le k \le r$,

$$\frac{g_k}{g_{k-1}} = \frac{1-z^{r-k+1}}{1-z^k}$$

is strictly decreasing in k, because its numerator decreases strictly and its positive denominator increases strictly. Hence $g_k^2 > g_{k-1}g_{k+1}$, that is, $\pi_k^2 > \pi_{k-1}\pi_{k+1}$, for $1 \le k \le r-1$. Multiplying the two inequalities between positive numbers gives $y_k^2 > y_{k-1}y_{k+1}$. □

Proof of Corollary 1.3. By Theorem 1.1 all coefficients $y_0, \dots, y_r$ are positive, so there are no internal zeros, and they are log-concave. □

Example 6.1. Table 1 lists $Y_{B_{r,s}(q)}$ and the normalized sequence F for $q \in \{2,3\}$ and $r \le 4$. For $(r,s,q) = (3,2,2)$ the matroid is $\mathrm{AG}(2,2) = U_{3,4}$, and the row $3,8,8,3$ agrees with [10, Theorem 1.1]; for $(r,s,q) = (3,1,2)$ it is the Fano plane with one point removed, that is, the cycle matroid of K_4.

TABLE 1. The coefficients y_k of $Y_{B_{r,s}(q)}$ and the values F_k.

q	r	s	$y_0, \dots, y_r$	$F_0, \dots, F_r$
2	2	1	$1, 2, 1$	1/2, 2/3, 1/2
2	3	1	$6, 12, 12, 6$	3/4, 6/7, 6/7, 3/4
2	3	2	$3, 8, 8, 3$	3/8, 4/7, 4/7, 3/8
2	4	1	$56, 112, 133, 112, 56$	7/8, 14/15, 19/20, 14/15, 7/8
2	4	2	$42, 96, 118, 96, 42$	21/32, 4/5, 59/70, 4/5, 21/32
2	4	3	$21, 64, 84, 64, 21$	21/64, 8/15, 3/5, 8/15, 21/64
3	2	1	$2, 3, 2$	2/3, 3/4, 2/3
3	3	1	$24, 36, 36, 24$	8/9, 12/13, 12/13, 8/9
3	3	2	$16, 27, 27, 16$	16/27, 9/13, 9/13, 16/27
3	4	1	$702, 1053, 1144, 1053, 702$	26/27, 39/40, 44/45, 39/40, 26/27
3	4	2	$624, 972, 1065, 972, 624$	208/243, 9/10, 71/78, 9/10, 208/243
3	4	3	$416, 729, 819, 729, 416$	416/729, 27/40, 7/10, 27/40, 416/729

Remark 6.2. The proofs above are complete and do not depend on computation. As an independent check, we computed $Y_{B_{r,s}(q)}$ in exact integer arithmetic for $q \in \{2,3,4,5,7,8,9,11,16,25\}$, $2 \le r \le 22$ and $1 \le s \le r-1$, that is, for 2310 triples (q,r,s). No subspaces are enumerated: the nonempty flats are grouped by their type $(d,a) = (\dim U, \dim(U \cap W))$, which occurs with the multiplicity $N(d,a)$ of (2.7); the restriction to a flat of type (d,a) is $B_{d,a}(q)$, and the contraction by a nonempty flat is a projective geometry (Lemma 2.3). With these data, $Y_{B_{r,s}(q)}$ is computed once from (1.1), with the inverse Kazhdan–Lusztig polynomials obtained by inverting P in the incidence algebra and with the Möbius invariants computed recursively from the same types, and once from Proposition 3.2. The two results agree in all cases. Moreover the endpoint formula of Theorem 1.2(i) holds in all 2310 cases (the program checks F_0 and the symmetry of the coefficients), the expansion of Theorem 1.2(ii) holds at all 33110 internal indices, the inequalities of Theorems 1.1 and 1.2 hold, and the formula of Example 3.3 holds in all 210 cases with $s = 1$. The program `verify_coefficients.py`, its output `exact_audit_results.json` and instructions for running them are provided as supplementary material.

USE OF AI-ASSISTED TOOLS

AI-assisted tools were used in the research and preparation of this manuscript.

References

[1] G. E. Andrews, *The Theory of Partitions*, Encyclopedia of Mathematics and its Applications, vol. 2, Addison-Wesley, Reading, MA, 1976.

[2] R. C. Bose and R. C. Burton, *A characterization of flat spaces in a finite geometry and the uniqueness of the Hamming and the MacDonald codes*, J. Combinatorial Theory **1** (1966), 96–104.

[3] T. Braden, L. Ferroni, J. P. Matherne, and N. Nepal, *Inverse Kazhdan–Lusztig polynomials of matroids under deletion*, arXiv:2510.01086.

[4] T. Braden, J. Huh, J. P. Matherne, N. Proudfoot, and B. Wang, *Singular Hodge theory for combinatorial geometries*, arXiv:2010.06088.

[5] R. Cheng and S. Liu, *Kazhdan–Lusztig polynomials of matroids need not be unimodal*, arXiv:2607.24186.

[6] B. Elias, N. Proudfoot, and M. Wakefield, *The Kazhdan–Lusztig polynomial of a matroid*, Adv. Math. **299** (2016), 36–70.

[7] L. Ferroni, J. P. Matherne, M. Stevens, and L. Vecchi, *Hilbert–Poincaré series of matroid Chow rings and intersection cohomology*, arXiv:2212.03190.

[8] A. L. L. Gao, Y.-X. Li, and Y. Li, *Inverse Kazhdan–Lusztig polynomials of fan matroids*, arXiv:2603.25254.

[9] A. L. L. Gao, Y. Li, and M. H. Y. Xie, *Equivariant inverse Z-polynomials of matroids*, arXiv:2601.17314.

[10] A. L. L. Gao, X. Ruan, and M. H. Y. Xie, *The inverse Z-polynomial of a matroid*, arXiv:2507.01332.

[11] A. L. L. Gao and M. H. Y. Xie, *The inverse Kazhdan–Lusztig polynomial of a matroid*, J. Combin. Theory Ser. B **151** (2021), 375–392.

[12] N. Proudfoot, Y. Xu, and B. Young, *The Z-polynomial of a matroid*, Electron. J. Combin. **25** (2018), no. 1, Paper No. 1.26.

[13] Rethlas, *Unimodality of inverse Z-polynomials of matroids*, manuscript, 2026, `https://github.com/frenzymath/Rethlas_results/blob/895e3d12ba02224c9d8ba8d52882184c77afc8af/AlgComb/MatroidKazhdanLusztig/inverse_z_unimodal.pdf`.

Email address: `math@knowledgesense.jp`